

Day 3 - Random Variables and Distributions

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Table of contents

From Data to Randomness	1
Discrete Random Variables	2
A Loaded Die	2
Continuous Random Variables	3
The Uniform Distribution	3
Other Continuous Distributions	4
Independence	6
Discrete Case	6
Continuous Case	7
Multiple Independent Measurements	8
Summary	8
Looking Ahead	8

From Data to Randomness

In the previous lectures, we treated our measurements as fixed numbers and asked how to estimate an unknown quantity from them. But where does the variation in our data come from?

In the gravitational acceleration experiment, each measurement differed slightly from the true value due to random errors—tiny fluctuations in equipment, air currents, timing, etc. To understand *why* certain estimators (like the mean) work better than others, we need to model this randomness mathematically.

Today we develop the language of **random variables** and **distributions**. As before, you'll discover the key ideas through exercises.

Discrete Random Variables

Let's start with a familiar example.

A Loaded Die

Suppose you have a six-sided die that's been tampered with. After rolling it many times, you record the following frequencies:



The outcome of rolling this die is a **random variable**—a quantity whose value is determined by chance. We typically denote random variables by capital letters like X .

A random variable is completely described by its **distribution**: the list of possible values and their probabilities.

Exercises: Discrete Distributions

Exercise 1: Let X be the outcome of rolling the loaded die above.

- (a) Write down $P(X = k)$ for each $k \in \{1, 2, 3, 4, 5, 6\}$.
- (b) Verify that $\sum_{k=1}^6 P(X = k) = 1$. Why must this be true?
- (c) Compute $P(X \leq 3)$ and $P(X > 4)$.

Exercise 2: You flip a fair coin 3 times. Let Y be the number of heads.

- (a) List all 8 possible outcomes (HHH, HHT, ...).
- (b) Find $P(Y = k)$ for $k = 0, 1, 2, 3$.
- (c) What is $P(Y \geq 2)$?

Exercise 3: Let X be a random variable taking values $1, 2, \dots, n$ with $P(X = k) = \frac{k}{c}$ for some constant c .

- (a) Find c in terms of n .
- (b) Compute $P(X \leq n/2)$ when $n = 10$.

Continuous Random Variables

Discrete random variables take values in a finite (or countable) set. But many quantities—like measurement errors, waiting times, or positions—can take any real value in some interval.

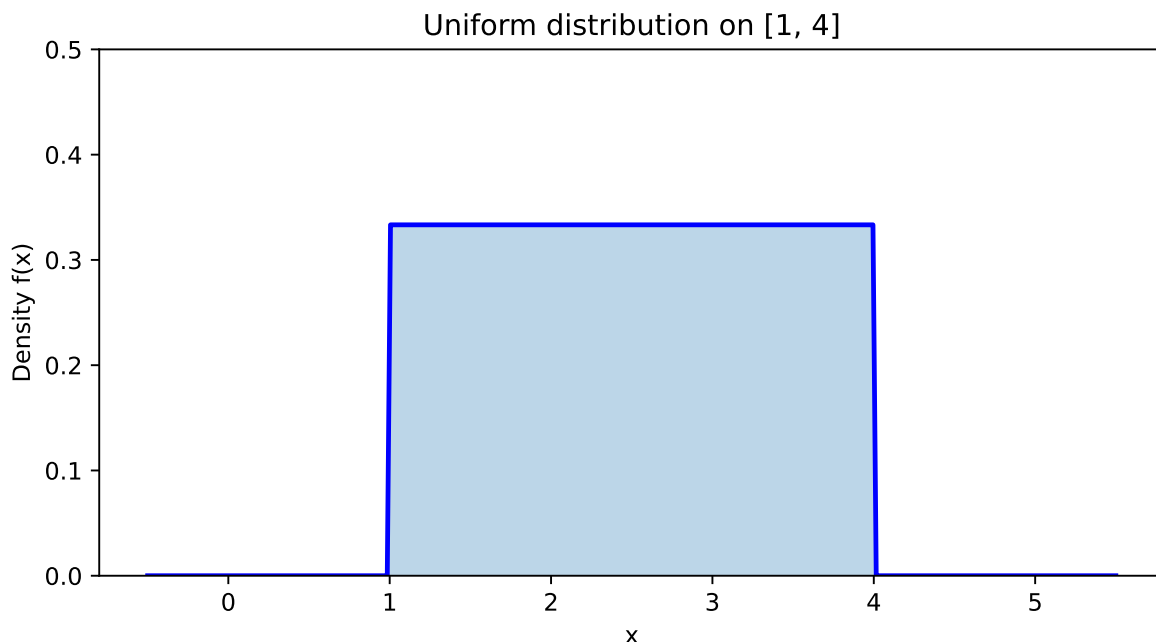
The Uniform Distribution

The simplest continuous distribution is the **uniform distribution** on an interval $[a, b]$, where every value is “equally likely.”

But what does “equally likely” mean when there are infinitely many values? We can’t assign a positive probability to each point (they’d sum to infinity). Instead, we describe probabilities using a **probability density function (pdf)**.

For the uniform distribution on $[a, b]$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



The key idea: **probability = area under the density curve.**

$$P(c \leq X \leq d) = \int_c^d f(x) dx$$

i Exercises: Uniform Distribution

Exercise 4: Let X be uniformly distributed on $[0, 1]$.

- (a) What is $P(X \leq 0.5)$?
- (b) What is $P(0.25 \leq X \leq 0.75)$?
- (c) What is $P(X = 0.5)$? (Hint: what's the area of a line segment?)

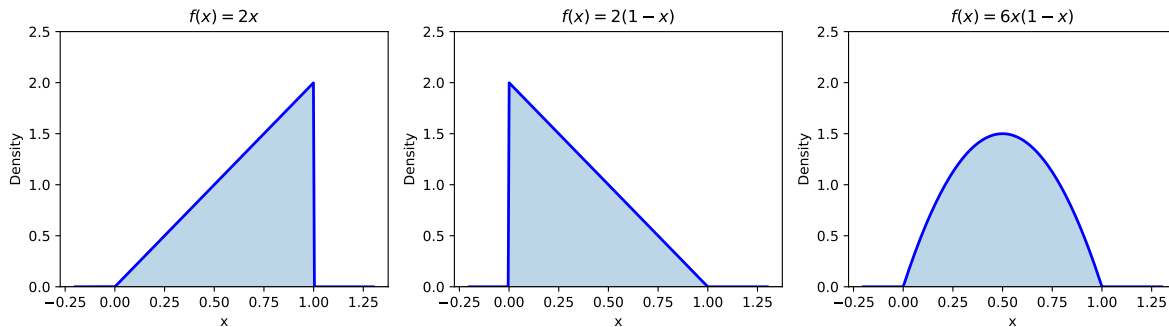
Exercise 5: Let X be uniformly distributed on $[a, b]$.

- (a) Verify that $\int_{-\infty}^{\infty} f(x) dx = 1$.
- (b) Compute $P(X \leq \frac{a+b}{2})$.

Other Continuous Distributions

What if some values are more likely than others? Consider a density that increases linearly:

$$f(x) = \begin{cases} cx & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



i Exercises: Other Densities

Exercise 6: Let X have density $f(x) = cx$ on $[0, 1]$ (and 0 elsewhere).

- Find the constant c that makes this a valid pdf.
- Compute $P(X \leq 0.5)$.
- Compute $P(X \geq 0.5)$. Verify that your answers to (b) and (c) sum to 1.

Exercise 7: Let X have density $f(x) = c(1 - x)$ on $[0, 1]$.

- Find c .
- Compute $P(X \leq 0.5)$.
- Compare with Exercise 6. What's the relationship between these two distributions?

Exercise 8: Let X have density $f(x) = c \cdot x(1 - x)$ on $[0, 1]$.

- Find c .
- Where is the density maximized?
- Compute $P(0.25 \leq X \leq 0.75)$.

Exercise 9: Let X have density $f(x) = c \cdot x^2$ on $[0, 1]$.

- Find c .
- Compute $P(X \leq 0.5)$.
- Compare this to your answer in Exercise 6(b). Which distribution puts more weight near $x = 1$?

💡 The Beta Family

The densities $f(x) \propto x^a(1-x)^b$ on $[0, 1]$ form a family called the **Beta distributions**. They're useful for modeling random quantities that are bounded between 0 and 1, like probabilities themselves. The exercises above correspond to:

- Exercise 6: Beta(2, 1)
- Exercise 7: Beta(1, 2)
- Exercise 8: Beta(2, 2)
- Exercise 9: Beta(3, 1)

Independence

When we take multiple measurements, we usually assume they don't influence each other—knowing that the first measurement was 9.82 m/s² doesn't change the probabilities for the second measurement. This concept is called **independence**.

Discrete Case

i Exercises: Discovering Independence

Exercise 10: Flip a fair coin twice. Let X be the first result (1 for heads, 0 for tails) and Y the second.

- List all four outcomes and their probabilities.
- Compute $P(X = 1)$ and $P(Y = 1)$.
- Compute $P(X = 1 \text{ and } Y = 1)$.
- Verify that $P(X = 1 \text{ and } Y = 1) = P(X = 1) \cdot P(Y = 1)$.
- Check that this relationship holds for all four combinations of values of X and Y .

Exercise 11: Roll a fair die twice. Let X be the first roll and Y the second.

- What is $P(X = 3 \text{ and } Y = 5)$?
- What is $P(X = 3) \cdot P(Y = 5)$?
- Are these equal? Is this what you'd expect?

Exercise 12: Draw two cards from a standard deck *without replacement*. Let $X = 1$ if the first card is an ace (0 otherwise), and $Y = 1$ if the second card is an ace.

- Compute $P(X = 1)$.
- Compute $P(X = 1 \text{ and } Y = 1)$. (Hint: use $P(A \text{ and } B) = P(A) \cdot P(B | A)$.)

- (c) Compute $P(X = 1) \cdot P(Y = 1)$.
- (d) Are (b) and (c) equal? Explain why this makes sense.

Exercise 13: Based on the previous exercises, propose a definition: when should we call two discrete random variables **independent**?

Two discrete random variables X and Y are **independent** if:

$$P(X = x \text{ and } Y = y) = P(X = x) \cdot P(Y = y) \quad \text{for all } x, y$$

In words: knowing the value of X tells you nothing about Y , and vice versa. The joint probability factors as a product of the individual probabilities.

Continuous Case

For continuous random variables, independence means the joint density factors:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

i Exercises: Independence for Continuous Variables

Exercise 14: Let X and Y be independent, both uniformly distributed on $[0, 1]$. The pair (X, Y) is a random point in the unit square.

- (a) What is $P(X \leq 0.5 \text{ and } Y \leq 0.5)$?
- (b) Sketch the region where $X \leq 0.5$ and $Y \leq 0.5$ in the unit square. Verify your answer to (a) by computing the area.

Exercise 15: Let X and Y be independent, both uniformly distributed on $[0, 1]$.

- (a) What is $P(X \leq 0.3 \text{ and } Y \geq 0.7)$?
- (b) What is $P(X \leq Y)$? (Hint: sketch the region in the unit square.)

Exercise 16: Let X and Y be independent, both with density $f(x) = 2x$ on $[0, 1]$.

- (a) What is the joint density $f_{X,Y}(x, y)$?
- (b) Compute $P(X \leq 0.5 \text{ and } Y \leq 0.5)$.

Exercise 17: (Challenge) Let X and Y be independent, both uniformly distributed on $[0, 1]$. Let $M = \max(X, Y)$.

- (a) Show that $P(M \leq t) = P(X \leq t \text{ and } Y \leq t) = t^2$ for $t \in [0, 1]$.
- (b) Find the density of M . (Hint: differentiate.)
- (c) What distribution from our earlier exercises does this match?

Multiple Independent Measurements

The power of independence becomes clear when we have many measurements.

i Exercises: Many Measurements

Exercise 18: Suppose you flip a fair coin 5 times. Let X_1, X_2, X_3, X_4, X_5 be the results (1 for heads, 0 for tails), and assume they are independent.

- (a) What is $P(X_1 = 1 \text{ and } X_2 = 1 \text{ and } X_3 = 1 \text{ and } X_4 = 1 \text{ and } X_5 = 1)$?
- (b) What is $P(X_1 = 0 \text{ and } X_2 = 1 \text{ and } X_3 = 0 \text{ and } X_4 = 1 \text{ and } X_5 = 0)$?
- (c) What is the probability of any specific sequence of 5 flips?

Exercise 19: Suppose you make 3 independent measurements X_1, X_2, X_3 , each uniformly distributed on $[0, 1]$.

- (a) What is $P(X_1 \leq 0.5 \text{ and } X_2 \leq 0.5 \text{ and } X_3 \leq 0.5)$?
- (b) What is $P(\text{all three measurements are } \leq 0.5)$?
- (c) What is $P(\text{at least one measurement is } > 0.5)$?

Exercise 20: Let $X_1, X_2, \dots, X_n$ be independent random variables, each with density $f(x) = 2x$ on $[0, 1]$.

- (a) Write down the joint density $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$.
- (b) Compute $P(X_1 \leq 0.5 \text{ and } X_2 \leq 0.5 \text{ and } \dots \text{ and } X_n \leq 0.5)$.

Summary

Concept	Discrete	Continuous
Distribution	Values and their probabilities	Probability density function $f(x)$
Probabilities	$P(X = k)$	$P(a \leq X \leq b) = \int_a^b f(x) dx$
Must satisfy	$\sum_k P(X = k) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Independence	$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$	$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$

Looking Ahead

We now have the vocabulary to model measurement error as a random variable. In the next lecture, we'll use these tools to answer the question from Day 2: *which estimator should we*

prefer?

The key idea will be **maximum likelihood**: given observed data, which value of the unknown parameter makes that data most probable? We'll discover that this question leads us to a very specific—and very famous—probability distribution.